

Aradhye Agarwal

aradhyeagarwal@gmail.com LinkedIn GitHub Google Scholar

EDUCATION

Indian Institute of Technology Delhi

Delhi, India

B.Tech + M.Tech in CS (Dual Degree) **GPA: 9.00/10 (M.Tech), 8.26/10 (B.Tech)**

Sep 2020 – Aug 2025

WORK EXPERIENCE

Research Fellow at Microsoft Research India – Reasoning Team

Jul 2025 – Mar 2026

- First-authored MOSAIC (ICML 2026): a post-training framework aligning agentic LMs for safe multi-step tool use via a plan-check-act/refuse loop, trained with pairwise preference RL across four model families
- Reduced harm score from 14% to 9% (-36% relative) on AgentHarm, with fine-tuned SLMs (4B–30B) outperforming unscaffolded GPT-4o/GPT-5 baselines while preserving task utility
- Co-advised by Akshay Nambi (MSR India) and Ahmed Awadallah (MSR AI Frontiers)

Software Engineering Intern at Google India – Cloud Team

May 2024 – Jul 2024

- Proposed and built a no-code meta-templating system where non-technical admins compose vendor-contract templates from typed, nestable fields – designed to replace a 5-day engineering cycle per template
- Implemented the full stack (TypeScript, Angular; 4,000+ LOC merged into Google’s monorepo), serving admin-generated templates at runtime alongside code-defined ones for backward compatibility; demoed end-to-end
- Received a full-time return offer

PUBLICATIONS

Learning When to Act or Refuse:

Guarding Agentic Reasoning Models for Safe Multi-Step Tool Use

International Conference on Machine Learning (ICML) 2026

Aradhye Agarwal, Gurdit Siyan, Yash Pandya, Joykirat Singh, Akshay Nambi, Ahmed Awadallah

Step-by-Step Unmasking for Parameter-Efficient Fine-tuning of Large Language Models

Transactions of the Association for Computational Linguistics (TACL) 2024

Aradhye Agarwal*, Suhas K. R.*, Ayan Sengupta*, Tanmoy Chakraborty

ECoDepth: Effective Conditioning of Diffusion Models for Monocular Depth Estimation

Computer Vision and Pattern Recognition (CVPR) 2024 · 120+ citations

[Code]

Suraj Patni*, **Aradhye Agarwal***, Chetan Arora

The Third Monocular Depth Estimation Challenge

Computer Vision and Pattern Recognition Workshop (CVPRW) 2024

Jamie Spencer, [...], **Aradhye Agarwal**, [...], James H. Elder

The Art of Scaling Test-Time Compute for Large Language Models

arXiv preprint, 2025

Aradhye Agarwal, Ayan Sengupta, Tanmoy Chakraborty

*Equal contribution

ACHIEVEMENTS

Won USD 10,000 in Kaggle’s “Child Mind Institute – Problematic Internet Usage,” placing 2 of 3560 teams *Dec 2024*

One of 50 students from across India invited to attend the Google DeepMind Research Symposium *Jan 2025*

One of the winners of The Third Monocular Depth Estimation Challenge (CVPR 2024 Workshop) *Jun 2024*

TALKS

ICML 2026 poster presentation of MOSAIC in Seoul

July 2026

The Third Monocular Depth Estimation Challenge: CVPR’24 Workshop Talk

Jun 2024

SKILLS

Post-training: PyTorch, TRL (GRPO, DPO), preference-based RL, PEFT/LoRA, diffusion models (stable diffusion)

Inference & systems: vLLM, test-time scaling, quantization, ONNX export, GCP (A100s), CUDA toolchain, Linux

Languages & data: Python, TypeScript, C++; synthetic speech-dialogue pipelines, audio codecs, LLM-judge evals